FocusBuddy

AI Feature and Responsible AI

1. AI Feature Overview

What the AI Feature Does

FocusBuddy's core AI feature converts unstructured, anxiety-laden user input into a sequenced set of concrete, ultra-low-friction physical micro-actions. When a user types or speaks a brain-dump of their stress — "I have no idea where to start with this assignment and I've been avoiding it for three days" — the AI strips the emotional content, identifies the underlying task, and returns a single, specific, immediately actionable first step. It dynamically adjusts the difficulty and tone of each step based on the user's self-reported cognitive state at the time of the session.

Where It Appears in the User Journey

The AI activates at the **moment of paralysis,** between the user's initial free-form input (the vent screen) and the action workspace (the step-by-step card view). It is the bridge that converts overwhelm into a starting point. It also re-engages at each step transition, recalibrating difficulty based on completion signals or user feedback.

Why AI Is Essential for This Problem

Hardcoded rule-based software requires structured input: specific task names, due dates, subtask categories. A user in executive dysfunction freeze **cannot produce structured input** - that is precisely the problem. Large language models are uniquely suited here because they can interpret vague, emotional, ungrammatical natural language and extract an actionable task context without requiring the user to do any cognitive heavy lifting.

2. Type of AI Capability Used

FocusBuddy combines several AI capabilities:

1. **Conversational AI** to understand natural language inputs.
2. **Summarization** to extract actionable tasks from unstructured thoughts.
3. **Classification** to identify user context such as overwhelm, avoidance, or low energy.
4. **Personalization Engine** to adapt guidance and step size based on user needs.
5. **AI Agent Workflow** to continuously guide users through task initiation and progression.

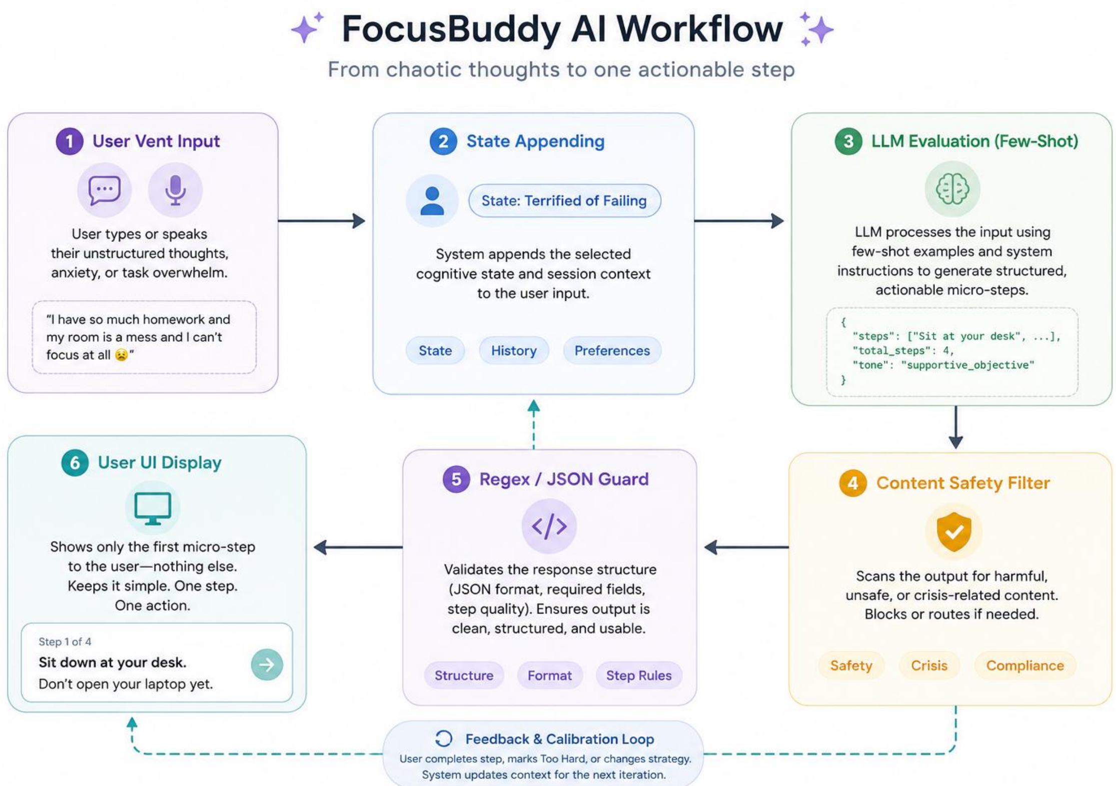
3. Input and Output Matrix

INPUT GIVEN TO AI	AI PROCESSING / LOGIC	OUTPUT SHOWN TO USER
Free-form anxiety text + cognitive state toggle (e.g. "Terrified of Failing")	Identifies the core task buried in emotional language. Classifies task type and complexity. Generates a sequenced micro-action array calibrated to the selected state.	"Step 1 of 4: Sit down at your desk and open a blank document. Don't write anything yet."
"Too Hard" feedback tap on current step	Receives negative signal on current step difficulty. Retrieves the same task context. Re-generates a simpler physical sub-action that precedes the current step.	"Let's go smaller. Step 1 of 5: Just find the document on your computer and open it. Nothing else."
Step completion tap ("Done")	Logs completion signal. Advances array index. Slightly raises the complexity threshold for the next step based on momentum signal.	"Great. Step 2 of 4: Read only the first paragraph of the brief. Just read — no notes yet."
Re-entry after session abandonment	Detects incomplete session. Retrieves last known task context and step position. Generates a low-friction re-entry prompt without shame or streak logic.	"Welcome back. No pressure — you were working on your thesis intro. Want to pick up where you left off, or start fresh?"

4. AI Workflow Step-by-Step

1. **Ingestion:** The user submits unstructured text or speech to the zero-friction input arena.
2. **Context Assembly:** The system pairs the raw text with active frontend state variables (e.g., chosen strategy toggle, historical step-calibration modifiers).
3. **Prompt Execution:** The grouped payload enters the LLM API container utilizing a rigid system prompt that mandates strict structural execution and output format rules.
4. **Content Filtering (Pre-Output Guardrails):** The generated token stream passes through automated validation hooks to scan for system hallucinations, safety breaches, or syntax formatting errors.

5. **Progressive Render:** The validated, structured JSON parses down to the user interface, revealing **only** the very first array index point to the user.
6. **Feedback Calibration Loop:** The user acts on the prompt via a completion click or requests a micro-downscale nudge, immediately updating the state payload for the subsequent interaction.



5. Data Architecture & Sensitive Information Boundaries

Data Utilized by the System

1. **User-Provided Operational Data:** Unstructured situational descriptions typed or spoken during task-rescue events.
2. **Volitional Context Metadata:** Selected behavioral states (*Overwhelmed, Avoiding, Low Energy*).
3. **Session Progress Metrics:** Completion velocity rates and explicit step adjustment selections ("*Too Hard*" clicks).

Sensitive Data Policy & Compliance

Because executive dysfunction relates directly to conditions like ADHD, autism, anxiety, and burnout, user inputs are strictly handled under a **Sensitive Data Protocol**:

1. **No Diagnostic Inferences:** The system strictly analyzes inputs for immediate task-initiation context and **never** deduces, logs, or tracks clinical conditions, psychiatric diagnoses, or medical profiles.
2. **Privacy-First Exclusions:** All connections to upstream LLM API endpoints utilize enterprise data privacy flags, ensuring user inputs are explicitly zero-retained by the provider and never utilized for model training loops.

6. Possible AI Failure Cases

1. **Macro-Step Hallucination (Failure to Decompose)**
   The AI misjudges task complexity and generates an initial step that remains too broad or abstract (e.g., "*Step 1: Write the entire executive summary*"), completely defeating the product's core intent.
2. **Contextual Misunderstanding**
   The user types an emotional metaphor (e.g., "*I'm drowning under a mountain of paper*"), and the AI literalizes the phrase, outputting steps related to physical safety or waste disposal instead of academic organization.
3. **Toxic Positivity or Patronizing Persona**
   The model generates overly enthusiastic, cheesy praise that violates the user's emotional safety boundaries, triggering defensive demand avoidance.
4. **API Response Latency**
   The underlying LLM infrastructure experiences high load, pushing token generation response times beyond 5.0 seconds, causing an anxious user to abandon the app mid-crisis.
5. **Hallucinated task requirements**
   The AI injects fabricated structural rules into the step breakdown — e.g. inventing a word count requirement or a submission format that does not exist — leading the user to do unnecessary or incorrect work.
6. **Cultural bias in step generation**
   A model trained predominantly on Western productivity frameworks may generate contextually inappropriate steps for Indian users — e.g. "email your professor" in an academic culture where in-person, hierarchical communication is the norm.

7. Responsible AI Risks

1. **Privacy Risk**
   Users offloading heavy anxiety may accidentally include high-risk Personally Identifiable

Information (PII) such as student IDs, corporate passwords, financial data, or workplace credentials within their text vent blocks.

2. **Bias & Fairness Risk**
   Cognitive styles differ across cultures, ages, and backgrounds. The AI could inherently favor specific Western, hyper-linear corporate productivity methodologies, alienating alternative processing archetypes.
3. **Hallucination Risk**
   The AI could inject fabricated structural rules, wrong instructions, or incorrect technical requirements into the step breakdowns it presents.
4. **User Trust & Safety Risk**
   A highly distressed user might leverage the vent container to express acute psychiatric crisis ideation or severe mental health emergencies that a utility app is unequipped to handle safely.
5. **Overdependence Loop**
   Users could become behaviorally dependent on the AI to initiate literally every action in their day, atrophying their innate self-regulation strategies over time.

8. Guardrails and Mitigations Matrix

RISK	POSSIBLE IMPACT	MITIGATION / GUARDRAIL
Safety — mental health crisis	User shares self-harm ideation or acute psychiatric distress the product cannot safely handle.	System prompt safety intercept: threshold keywords trigger a clean AI loop exit and surface a static crisis helpline modal (iCall, Vandrevala Foundation). No AI response is generated.
Privacy — PII leakage	Accidental exposure of credentials, IDs, or personal records via the vent input field.	Client-side regex scrubbing layer sanitises emails, phone numbers, numeric codes, and credential patterns before any string is dispatched to the cloud API.
Step size miscalibration	Overly broad first step re-triggers paralysis — defeating the product's core purpose.	Few-shot JSON schema enforcement: every step must begin with a physical action verb from a constrained list. Maximum complexity ceiling enforced per step. "Too Hard" tap triggers immediate downscale re-generation.
Overdependence	User loses autonomous initiation ability over time.	Progressive autonomy prompts: as session momentum builds, AI shifts from directive

		("Open the document") to open-ended ("What feels like the right next step?"), gradually returning agency to the user.
Bias & fairness added	Culturally misaligned step suggestions alienate Indian users or produce unhelpful outputs.	Culturally diverse few-shot prompt templates calibrated for Indian academic and professional contexts. Regional tone review in prompt design. Ongoing bias audit against user feedback signals.
User trust added	Confidently wrong AI output followed by user leads to wasted effort and permanent trust loss.	All step outputs framed as suggestions, not instructions: "Based on what you shared, try this..." Binary feedback on every card. Repeated negative feedback triggers a session reset prompt rather than continued generation.

9. AI Transparency & Communication Strategy

1. **Clear AI Labeling**
   The workspace utilizes prominent, persistent visual messaging stating: "*FocusBuddy is an automated AI co-pilot, not a human or a clinical tool.*"
2. **Explicit Onboarding Disclaimer**
   Before their very first session, users must clear a mandatory modal stating that the system is built purely as a behavioral action utility, is completely non-diagnostic, and does not store personal psychological histories.
3. **Active Feedback Signals**
   Every action card features inline binary micro-feedback options ([👍 Match] / [👎 Too Complex]). This empowers the user with immediate oversight and flags to the model whether its step calibration hit or missed the mark.
4. **Confidence indicators**
   Where the AI has low contextual confidence — e.g. the vent input was very short or ambiguous — the step card displays a soft indicator: "I had limited context for this — does this feel right?" prompting the user to add more details or regenerate.